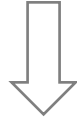


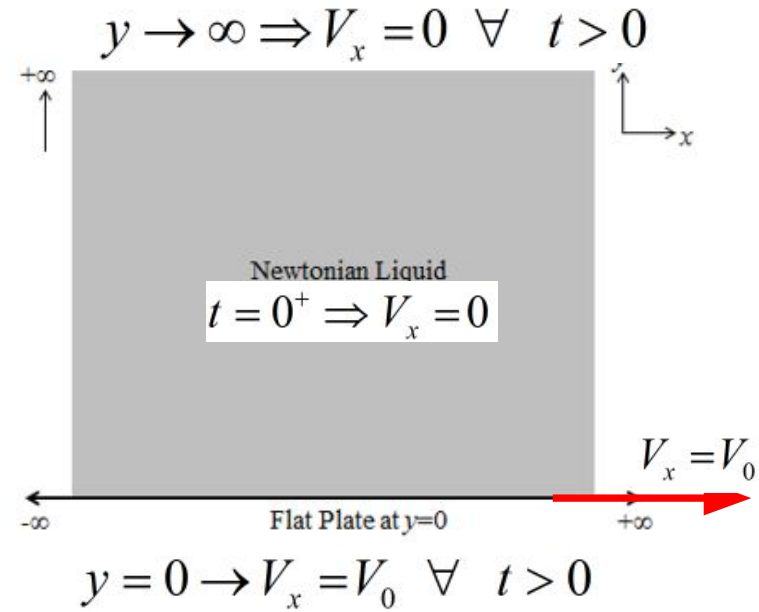
Stoke's 1st Problem: Flat-Plate being pulled with Constant Velocity

$$\left(\frac{\partial V_x}{\partial t} \right) = \frac{\mu}{\rho} \left(\frac{\partial^2 V_x}{\partial y^2} \right)$$



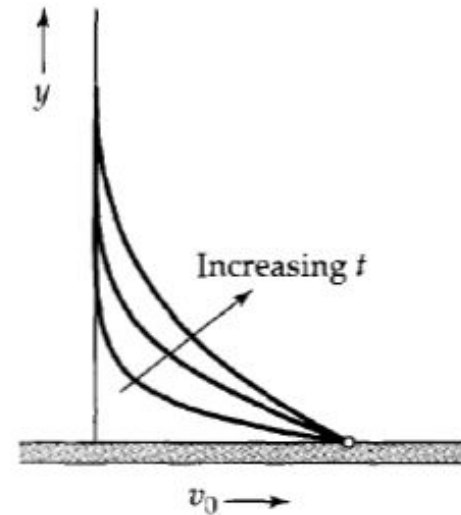
$$\frac{\partial V_x}{\partial t} = \nu \frac{\partial}{\partial y} \left(\frac{\partial V_x}{\partial y} \right)$$

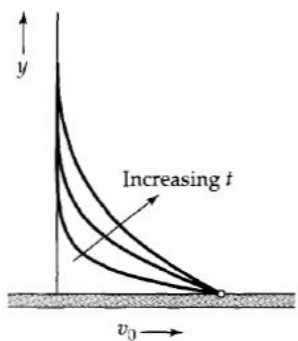
Momentum
Diffusivity



$$V_y \text{ \& } V_z = 0$$

$$\frac{\partial V_x}{\partial t} = \nu \frac{\partial}{\partial y} \left(\frac{\partial V_x}{\partial y} \right)$$





$$\left(\frac{\partial V_x}{\partial t}\right) = \frac{\mu}{\rho} \left(\frac{\partial^2 V_x}{\partial y^2}\right) \Rightarrow \boxed{\frac{\partial V_x}{\partial t} = \nu \left(\frac{\partial^2 V_x}{\partial y^2}\right)} \Rightarrow \left(V_0 \frac{\partial f}{\partial \eta}\right) \left(-\frac{\eta \delta'}{\delta}\right) = \nu \left(V_0 \frac{\partial^2 f}{\partial \eta^2} \frac{1}{\delta^2}\right) \Rightarrow \boxed{\frac{\frac{\partial^2 f}{\partial \eta^2}}{\eta \frac{\partial f}{\partial \eta}} = -\frac{\delta \delta'}{\nu} = c}$$

$$f = \frac{V_x}{V_0} = f(\eta), \quad \eta = \frac{y}{\delta(t)} \quad \frac{\partial \eta}{\partial t} = -\frac{y}{\delta^2} \frac{\partial \delta}{\partial t} = -\frac{y \delta'}{\delta^2} = -\frac{\eta \delta'}{\delta}$$

$$-\frac{\delta \delta'}{\nu} = -2 \Rightarrow \frac{\delta}{\nu} \frac{d\delta}{dt} = 2 \Rightarrow \frac{\delta^2}{2\nu} = 2t + c_4$$

$$t \rightarrow 0 \Rightarrow \delta \rightarrow 0 \Rightarrow c_4 = 0$$

$$\frac{\partial V_x}{\partial t} = \frac{\partial V_x}{\partial \eta} \frac{\partial \eta}{\partial t} = \left(V_0 \frac{\partial f}{\partial \eta}\right) \frac{\partial \eta}{\partial t} = \left(V_0 \frac{\partial f}{\partial \eta}\right) \left(-\frac{\eta \delta'}{\delta}\right)$$

$$\frac{\partial V_x}{\partial y} = \frac{\partial V_x}{\partial \eta} \frac{\partial \eta}{\partial y} = \left(V_0 \frac{\partial f}{\partial \eta}\right) \frac{1}{\delta(t)} \Rightarrow \frac{\partial^2 V_x}{\partial y^2} = V_0 \frac{\partial^2 f}{\partial \eta^2} \frac{1}{\delta^2}$$

$$\frac{\partial^2 V_x}{\partial y^2} = \frac{\partial \left(\frac{\partial V_x}{\partial y}\right)}{\partial y} = \frac{\partial \left(\frac{\partial V_x}{\partial y}\right)}{\partial \eta} \frac{\partial \eta}{\partial y} = \left(V_0 \frac{\partial^2 f}{\partial \eta^2} \frac{1}{\delta(t)}\right) \frac{1}{\delta(t)}$$

Order of magnitude analysis

$$\frac{V_0}{t_{ref}} \frac{\partial f}{\partial t^*} = \frac{\nu V_0}{\delta^2} \left(\frac{\partial^2 f}{\partial \eta^2}\right)$$

$$\frac{V_0}{t_{ref}} \sim \frac{\nu V_0}{\delta^2} \Rightarrow \frac{V_0}{t} \sim \frac{\nu V_0}{\delta^2} \Rightarrow \delta \sim \sqrt{\nu t}$$

Momentum
Diffusion length scale

$$\Rightarrow \boxed{\delta = \sqrt{4\nu t}}$$

$$\eta = \frac{y}{\delta(t)} = \frac{y}{\sqrt{4\nu t}}$$

$$\frac{\partial f}{\partial \eta} = g$$

$$\frac{\frac{\partial^2 f}{\partial \eta^2}}{\eta \frac{\partial f}{\partial \eta}} = -2 \Rightarrow \boxed{\left(\frac{dg}{d\eta} \right) \frac{1}{g}} = -2\eta \Rightarrow \frac{dg}{g} = -2\eta d\eta$$

$$\ln g = -\eta^2 + c_1 \Rightarrow g = e^{-\eta^2 + c_1} \Rightarrow g = e^{c_1} e^{-\eta^2} \Rightarrow g = c_2 e^{-\eta^2}$$

$$y \rightarrow 0 \Rightarrow \eta \rightarrow 0, \quad f|_{\eta=0} = \frac{V_x}{V_0} = \frac{V_0}{V_0} = 1$$

$$\frac{df}{d\eta} = c_2 e^{-\eta^2} \Rightarrow f|_{\eta} - f|_{\eta=0} = \int_0^{\eta} c_2 e^{-\eta^2} d\eta \Rightarrow f(\eta) = 1 + \int_0^{\eta} c_2 e^{-\eta^2} d\eta \Rightarrow f|_{\eta=\infty} = 0 = 1 + c_2 \int_0^{\infty} e^{-\eta^2} d\eta$$

$$t \rightarrow 0 \text{ or } y \rightarrow \infty \Rightarrow \eta \rightarrow \infty,$$

Initial
condition
n

Boundary
condition

$$f|_{\eta=\infty} = \frac{V_x}{V_0} = \frac{0}{V_0} = 0$$

$$c_2 = \frac{-1}{\int_0^{\infty} e^{-\eta^2} d\eta}$$

$$c_2 = \frac{-1}{\int_0^{\infty} e^{-\eta^2} d\eta} = \frac{-2}{\int_0^{\infty} z^{-1/2} e^{-z} dz} = \frac{-2}{\int_0^{\infty} z^{(1/2)-1} e^{-z} dz} = \frac{-2}{\Gamma(1/2)} = \frac{-2}{\sqrt{\pi}}$$

$$f(\eta) = 1 - \left(\frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-\eta^2} d\eta \right) = 1 - \operatorname{erf}(\eta)$$

$$f(\eta) = 1 - \operatorname{erf}(\eta)$$

$$\boxed{\frac{V_x(y,t)}{V_0} = 1 - \operatorname{erf}\left(\frac{y}{\sqrt{4\nu t}}\right)}$$

$$\left(\frac{\partial V_x}{\partial t}\right) = \frac{\mu}{\rho} \left(\frac{\partial^2 V_x}{\partial y^2}\right)$$

$$t = 0^+ \Rightarrow V_x = 0$$

$$y \rightarrow \infty \Rightarrow V_x = 0 \quad \forall \quad t > 0$$

$$y = 0 \rightarrow V_x = V_0 \quad \forall \quad t > 0$$